SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

I AGNEESHWARAN B(192524508) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AGNEESHWARAN B

Annexure A

<u>Constraint-Based Problem Solving</u>

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right
ii. Cost of each move = 1 iii. Cannot pass through obstacles iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ANSWERS

1. Constraint-Based Problem Solving (Backtracking Search)

Problem

Assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) with the following constraints:

- D1 cannot work Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor must be assigned exactly one shift.

Backtracking Search Steps

Step 1: Assign D1

Possible shifts for D1:

- Morning ✓
- Afternoon ✓
- Night ✗ (Not allowed)

Assign **D1 → Morning**.

Current Assignment:

- Morning → D1
- Afternoon → Empty
- Night → Empty

Step 2: Assign D2

Available shifts:

- Afternoon
- Night

Since D2 must work before D3, assign:

D2 → Afternoon

Current Assignment:

- Morning → D1

- Afternoon → D2
- Night → Empty

Step 3: Assign D3

Remaining shift:

- Night

Check constraints:

- D3 cannot work Morning ✓
- D2 works before D3 (Afternoon before Night) ✓
- One doctor per shift ✓

Assign:
D3 → Night

Constraint Checking

Constraint	**Status**
D1 not Night	✓ Satisfied
D2 before D3	✓ Satisfied
One doctor per shift	✓ Satisfied
D3 not Morning	✓ Satisfied
Every doctor assigned one shift	✓ Satisfied

Final Valid Schedule

Shift	**Doctor**
Morning	D1
Afternoon	D2
Night	D3

2. Breadth-First Search (BFS)

Given Constraints

- Grid size: 5 × 5
- Start = S
- Goal = G
- Obstacles block movement
- Allowed movements:
  - Up
  - Down
  - Left
  - Right
- Cost of each move = 1
- Cannot move through obstacles
- Manhattan Distance heuristic is available.

BFS Algorithm

1. Start from node **S**.
2. Add S to the queue.
3. Mark S as visited.
4. Remove the first node from the queue.
5. Expand all valid neighbouring nodes.
6. Ignore visited nodes and obstacles.
7. Add new nodes to the queue.
8. Repeat until Goal (G) is found.
9. Trace back the path from Goal to Start.

Step-by-Step Node Expansion

Step 1

Queue:
[S]

Visited:
S

Expand S.

Step 2

Add neighbouring cells.

Queue:
[A, B]

Visited:
S, A, B

Step 3

Expand A.

Queue:

[B, C, D]

Visited:
S, A, B, C, D

Step 4

Expand B.

Obstacle encountered.
Ignore obstacle.

Queue:
C, D, E

Visited:
S, A, B, C, D, E

Step 5

Continue expansion until Goal is reached.

Queue:
Goal

Priority Queue Updates
Since BFS uses a normal queue (FIFO), nodes are expanded in the order they are inserted.
Example:
Initial Queue:
S

↓

A B

↓

C D E

↓

Goal

Optimal Path
S → A → C → E → G

Path Cost
Each move costs **1**.
If there are **4 moves**:
Total Cost = 4

4. Autonomous Rescue Robot

a) Analyze the Constraints

1. Robot can move Up, Down, Left, Right

The robot is allowed to move only in four directions. Diagonal movement is not allowed.

2. Obstacles cannot be crossed
Blocked cells must be avoided while planning the path.

3. Limited sensing capability
The robot can observe only adjacent cells. It does not know the entire environment initially.

4. Risky zones
Risky cells can be crossed, but each risky move adds an extra cost of **2**.
Therefore,
Safe cell cost = 1
Risky cell cost = 3
The robot prefers safer paths whenever possible.

5. Minimize total cost
The robot selects the path with the lowest overall travel cost while safely reaching the survivor.

6. Dynamic knowledge update
As the robot moves, it discovers new information and updates its map continuously.

Effect on Robot Decision-Making
The robot:

- Avoids blocked cells.
- Chooses safe routes.
- Minimizes travel cost.
- Updates knowledge during movement.
- Changes the planned path if new obstacles are discovered.

b) Appropriate Search Strategy
Uniform Cost Search (UCS)
Uniform Cost Search is the most suitable algorithm because movement costs are different due to risky zones.

Steps
1. Start from S.
2. Insert S into the priority queue.
3. Expand the node with the lowest total cost.
4. Ignore blocked cells.
5. Calculate the new path cost.
6. Update the priority queue.
7. Continue until Goal (G) is reached.
8. Return the minimum-cost path.

c) AI Concepts Used
Intelligent Agent
The rescue robot senses the environment, processes available information, and takes actions to reach the survivor safely.

Rationality
The robot always selects the action that minimizes total cost while avoiding obstacles and risky zones.

Environment Type
The environment is:
- Partially Observable
- Dynamic
- Sequential
- Discrete
- Unknown

d) Justification
Uniform Cost Search is the best choice because:
- It always finds the minimum-cost path.
- It handles different movement costs effectively.
- It avoids risky paths when safer alternatives exist.
- It adapts well to dynamic environments.

- It ensures safe and efficient navigation to the survivor.